

Step 05 - Measure Wrongness: Error, Residual, Loss

A First Practical Machine Learning Workflow

What this document is for

Step 04 produced predictions for the held-back test rows. Step 05 turns those predictions into something inspectable: how far away they are from the actual prices.

By the end of this step, you should have:

- Compared actual and predicted prices row by row.
- Added residuals, absolute errors, and squared errors to the comparison table.

1. Where this step sits

The model has already been fitted. This step does not train a new model. It asks whether the predictions produced by that fitted model look useful.

Step	What changes
01 - Get the notebook ready	The environment and package checks exist.
02 - Create a tiny dataset	Rows, features, target, X, and y exist.
03 - Split before training	X_train, X_test, y_train, and y_test exist.
04 - Fit a regression model	A model learns from the training portion and predicts X_test.
05 - Measure wrongness	Predictions are compared with actual values using errors and loss.

The rule for this whole step

Evaluate predictions against the held-back answers. Do not refit the model in this step, and do not change the model just because one test row looks awkward.

2. Prediction is not evaluation

The comparison table from Step 04 is useful because it shows actual and predicted values together. But looking at two columns is still not a measurement. To evaluate a regression model, you need to measure the gaps between those two columns.

Thing	Plain-English meaning
actual value	The known answer from <code>y_test</code> .
prediction	The model's answer for the matching row in <code>X_test</code> .
residual	actual - predicted. It keeps the direction of the miss.
absolute error	The size of the miss, ignoring direction.
loss / metric	A single summary number made from many row-level errors.

Direction versus size

A residual tells you whether the model predicted too low or too high. An absolute error tells you how large the miss was. Both are useful, but they answer different questions.

3. Rebuild the comparison table with unrounded predictions

In Step 04, the displayed predictions were rounded so the table was easy to read. For measurement, keep the raw prediction values in the table and round only when displaying.

Why this cell exists

This cell creates a fresh comparison table from `y_test` and `house_predictions`. It keeps the actual answers and model predictions beside each other, one row at a time.

Code cell

```
comparison = pd.DataFrame({
    "actual_price_k": y_test,
    "predicted_price_k": house_predictions,
})

comparison.round(1)
```

What to look for

The table should have one row for each held-back test example. The actual and predicted values are both in thousands of pounds because the target column, `price_k`, used that unit.

4. Add residuals

A residual is the signed difference between the actual value and the prediction.

Why this cell exists

The sign matters. A positive residual means the model predicted too low. A negative residual means the model predicted too high.

Code cell

```
comparison["residual_k"] = (
    comparison["actual_price_k"]
    - comparison["predicted_price_k"]
)

comparison.round(1)
```

What to look for

Read one row carefully. If the actual value is 410 and the prediction is 390, the residual is +20: the model was 20 thousand pounds too low. If the residual is -20, the model was 20 thousand pounds too high.

Avoid this common mistake

Do not average residuals and assume that means the model is good. Positive and negative residuals can cancel each other out, hiding large mistakes.

5. Add absolute error

Absolute error removes the sign and keeps only the size of the miss.

Why this cell exists

This makes row-by-row wrongness easier to compare. A miss of -25 and a miss of +25 are different directions, but they are the same size.

Code cell

```
comparison["absolute_error_k"] = comparison["residual_k"].abs()  
  
comparison.round(1)
```

What to look for

absolute_error_k should never be negative. It is measured in the same unit as price_k, so an absolute error of 30 means 30 thousand pounds.

6. Add squared error

Squared error is another way to make misses positive. It also makes large misses count much more heavily than small misses.

Why this cell exists

You do not need squared error to understand this first model, but it is worth seeing because many regression methods are built around squared loss.

Code cell

```
comparison["squared_error"] = comparison["residual_k"] ** 2  
  
comparison.round(1)
```

What to look for

Squared error is no longer in thousands of pounds. Its unit is squared thousands of pounds, which is awkward to read. That is one reason mean absolute error is friendlier as a first metric.

7. Calculate mean absolute error

Mean absolute error, often shortened to MAE, is the average absolute miss across the rows being evaluated.

Import one regression metric only when it is needed

This is the first point where the notebook needs a regression metric from scikit-learn. The metric does not train the model; it only compares answers.

Code cell

```
from sklearn.metrics import mean_absolute_error
```

Calculate test MAE

Use `y_test` because those are the held-back actual values. Use `house_predictions` because those are the model predictions for the matching `X_test` rows.

Code cell

```
test_mae = mean_absolute_error(y_test, house_predictions)
print("Test MAE:", round(test_mae, 2), "thousand pounds")
```

What to look for

If the test MAE is 30, the model is off by about 30 thousand pounds on average across the held-back rows. With only four test rows, treat the number as a learning signal, not a reliable housing-market result.

8. Check the manual calculation

It is useful to verify that the metric is not magic. MAE is just the average of the `absolute_error_k` column.

Why this cell exists

This connects the library metric to the row-level table you can inspect.

Code cell

```
manual_mae = comparison["absolute_error_k"].mean()
print("Manual MAE:", round>manual_mae, 2))
print("sklearn MAE:", round(test_mae, 2))
```

What to look for

The two printed values should match apart from tiny floating-point differences. That is the point: the scikit-learn metric is a reliable helper for a calculation you now understand.

9. What loss means in this workflow

Loss is a general word for wrongness measured by a chosen rule. The rule might be absolute error, squared error, cross-entropy, or something else. Different tasks and models use different loss functions.

For now

Use residuals to see direction, absolute errors to see miss size, and MAE as the first readable test summary. You are ready for Step 06 when you can explain those three ideas without looking back at the code.

Terms to Say Out Loud

Step 05 - Measure Wrongness: Error, Residual, Loss

Retrieval target

You do not need a dictionary definition. You should be able to say each term in plain English and connect it to the notebook work.

Term	Say it out loud as...
prediction	The model's answer for a feature row.
actual value	The known answer used to check the prediction.
residual	actual - predicted; the signed miss.
error	How wrong a prediction is under a chosen rule.
absolute error	The size of the miss, ignoring direction.
squared error	The miss squared, which punishes larger misses more strongly.
mean absolute error (MAE)	The average absolute miss across the evaluated rows.
loss	A chosen rule for turning wrong predictions into a number.
metric	A summary measurement used to report model performance.